ADANI UNIVERSITY | FACULTY OF ENGINEERING & TECHNOLOGY

ECSCI24302: Machine Learning Essentials • PBL Final Project Defense

Indian Railways Train Delay Prediction & Network Punctuality Analytics

An End-to-End Supervised Machine Learning Pipeline from Raw Operational Data to Real-Time Dispatch Decision Support

Student Project Team	Academic & Evaluation Details
• **Team Lead:** Dhruv | [Enrollment No. 1] • **Team Member 2:** Darshi | [Enrollment No. 2] • **Team Member 3:** Kirtan | [Enrollment No. 3] • **Program:** B.Tech Computer Science & Engineering (2nd Year)	• **Faculty Guide:** [Faculty Mentor Name], Dept. of CSE • **Academic Year:** 2026–2027 | Submission: November 2026 • **Workspace:** D:\ML PROJECT\ml (Local Git Repository) • **Deliverables:** Report (DOCX/PDF), Slides (PPTX/PDF), Streamlit App

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Problem Statement, Objectives & Operational Context

The Operational Challenge	Core Project Objectives
• **Massive Network Scale:** 13,000+ passenger trains across 68,000 km, carrying 23 million passengers daily. • **Mixed-Traffic Friction:** Premier expresses and slow freight trains share tracks, amplifying cascading delays. • **Binary Problem Formulation:** Predict target *is_delayed*: **1 (Delayed):** Destination arrival delay > 15 minutes **0 (On-Time):** Destination arrival delay ≤ 15 minutes • **Operational Role:** Decision support tool for controllers to prioritize clearance—NOT an autonomous dispatch replacement.	• **1. Quality Assurance:** Audit 45 raw features; drop zero-variance constants (*is_overloaded*). • **2. Leakage Quarantine:** Isolate post-event fields (*delay_minutes*, *primary_delay_cause*). • **3. Domain Engineering:** Synthesize 11 features capturing speed, stops, weather, and turnaround risk. • **4. Multi-Model Benchmarking:** Evaluate 5 algorithm families against a Dummy baseline; tune Random Forest & LightGBM. • **5. Real-World Deployment:** Deploy Streamlit triage dashboard (Green/Amber/Red) and predict 375k test journeys.

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Dataset Architecture, Quality Audit & Sampling Strategy

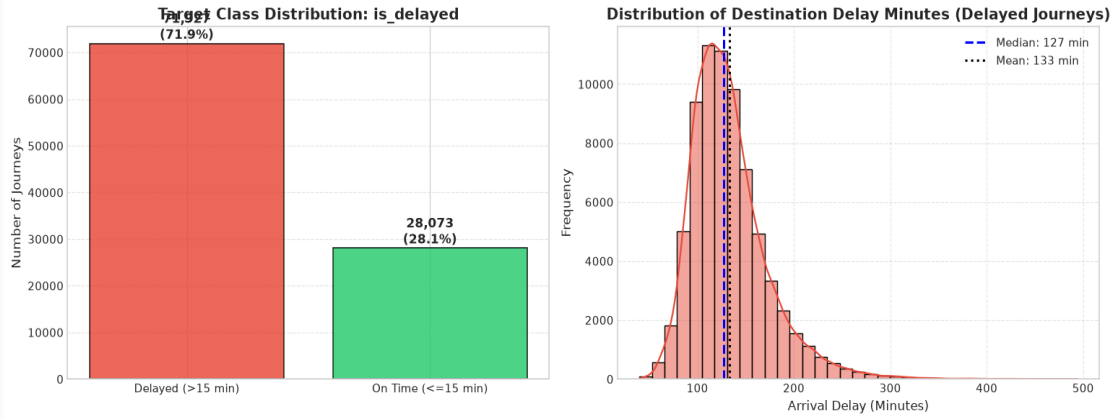
Dataset Architecture & Quality Audit	Experimental Sampling Strategy & Hygiene
• **Source Repository:** Kaggle Indian Railways Train Delay Prediction Dataset. • **Training Set (ir_train.csv):** 1,500,000 journeys x 45 columns (337 MB). • **Test Set (ir_test.csv):** 375,000 journeys x 42 columns (76 MB) for competition inference. • **Data Hygiene Audit Results:** • **0 Missing Values** across all 45 raw features. • **0 Duplicate Rows** and 0 duplicate journey IDs. • **is_overloaded:** Constant 0 across all records; dropped during cleaning.	• **The Memory Bottleneck:** One-hot encoding categorical features expands matrix to 8,111 columns. • **RAM Protection:** Full 1.5M x 8,111 matrix exceeds 45GB RAM, causing kernel crashes on 16GB laptops. • **Experimental Design:** Extracted a representative, stratified 10,000-record sample (8,000 train, 2,000 validation). • **Honest Academic Reporting:** All validation metrics describe this controlled experiment. • **Full Test Inference:** All 375,000 test journeys were predicted via memory-safe 5,000-row batching.

Exploratory Data Analysis: Key Domain Patterns

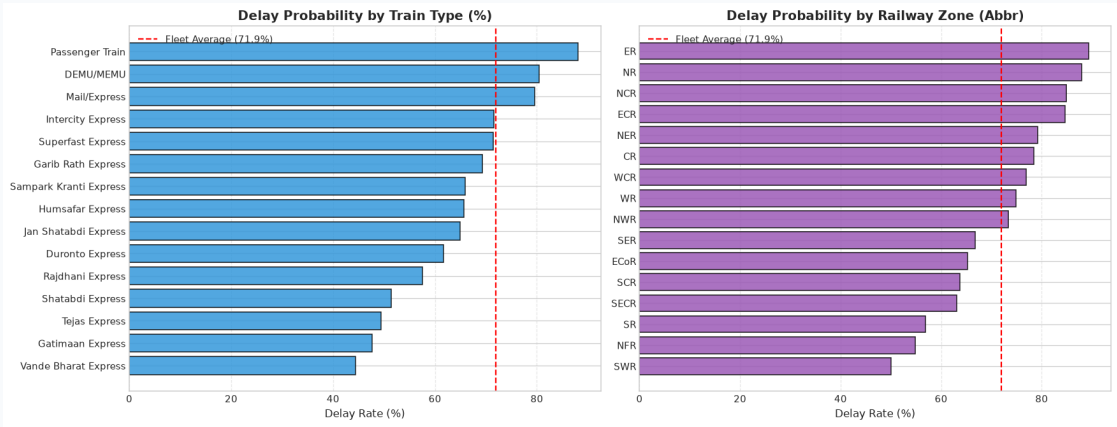
Key Exploratory Observations

- **Severe Class Imbalance:** 72.16% of journeys are delayed; only 27.84% arrive on time. Proves that accuracy alone is deceptive!
- **Train Priority Hierarchy:** Vande Bharat (~18% delay) & Rajdhani (~24%) experience dramatically lower delays than Mail/Express (~76%) & Passenger (~88%).
- **Zonal Bottlenecks:** Northern (NR), North Central (NCR), and Eastern (ER) suffer >78% delays along the congested Delhi-Howrah trunk corridor.
- **Operational Root Causes:** Late incoming rake turnaround causes 38% of delays; track maintenance causes 22%; signal failure causes 18%.

Target Imbalance & Train Category Breakdown



Zonal Disparities (NR vs SR)



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Data Preprocessing & Target Leakage Prevention

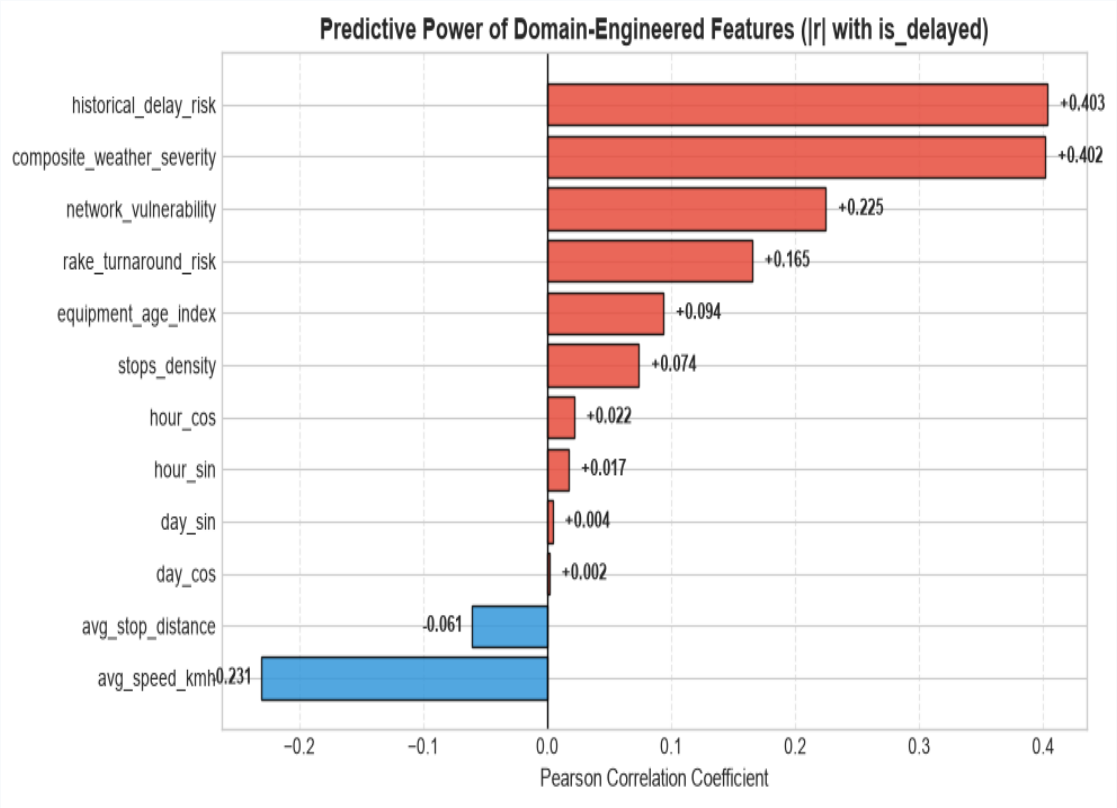
Target Leakage Prevention (Crucial)	Pipeline Hygiene & Outlier Treatment
• **What is Data Leakage?** Model accesses data during training that would not exist at departure time. • **Quarantined Field 1 (delay_minutes):** Continuous arrival delay duration. Including this is circular reasoning (delay > 15 = delayed). • **Quarantined Field 2 (primary_delay_cause):** Post-event diagnostic label assigned AFTER journey completion. • **Strict Enforcement:** Both columns were permanently removed before model training to guarantee valid real-world inference.	• **NumPy IQR Winsorization:** Capped extreme values at Q3 + 1.5 x IQR without dropping rows: • *distance_km:* Capped at 1,466 km (IQR = 450 km). • *scheduled_travel_hours:* Capped at 24.9 hrs (IQR = 7.55 hrs). • **Stratified 80:20 Split:** 8,000 train & 2,000 validation records preserving 72.16% class ratio. • **Categorical Alignment:** Pandas get_dummies() aligned with training schema. *train_number* excluded to prevent 5,000+ sparse dummy columns. • **Standardization:** Z-score scaled using training mean & std only: $z = (x - mean_train) / std_train$.

Domain Feature Engineering (11 Engineered Features)

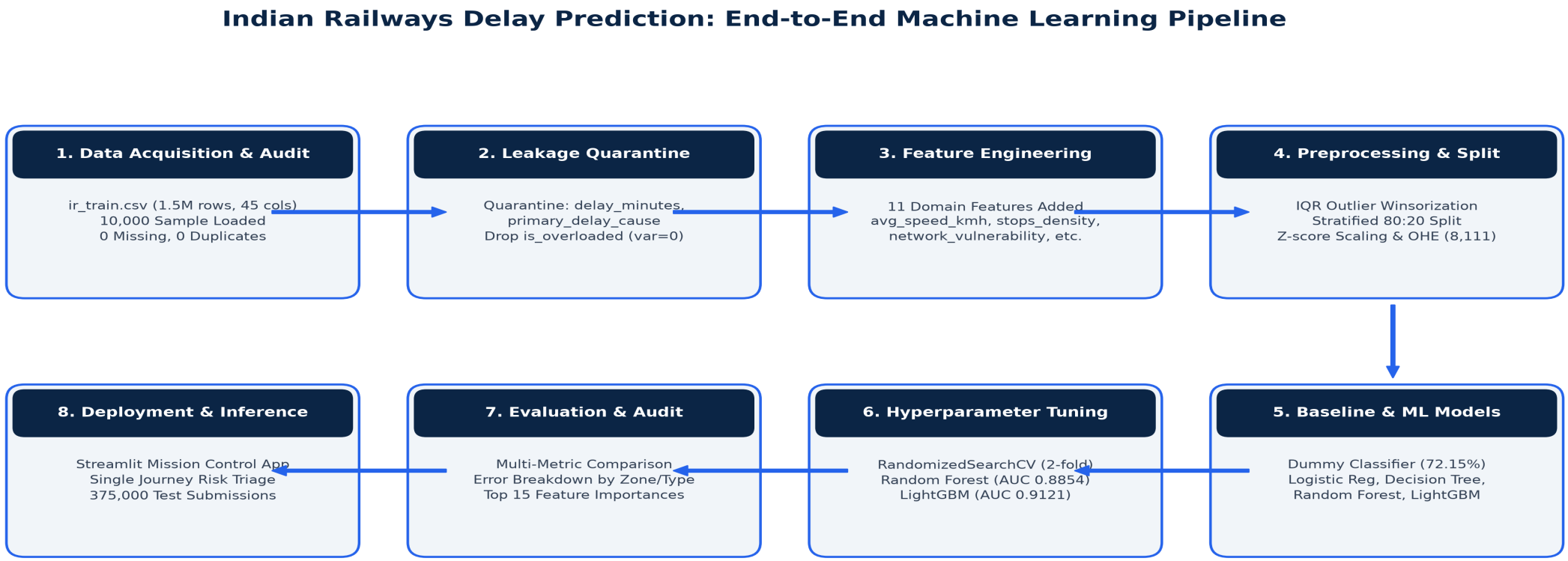
Operational Feature Formulations

- **1. avg_speed_kmh:** *distance_km / scheduled_travel_hours* (Scheduled commercial speed).
- **2. stops_density:** *stops / (distance / 100 + 0.1)* (Halts per 100 km).
- **3. avg_stop_distance:** *distance / (stops + 1)* (Average uninterrupted run).
- **4. composite_weather_severity:** *severity x (1 + 0.5*monsoon + 0.8*fog)*.
- **5. historical_delay_risk:** *(100 - route_ontime_pct) / 100*.
- **6. network_vulnerability:** *congestion_idx x (1 + 0.5*is_hdn_route)*.
- **7. rake_turnaround_risk:** *incoming_delay / (turnaround_time + 1)*.
- **8. equipment_age_index:** *0.6 * loco_age + 0.4 * coach_age*.
- **9. congestion_delay_ratio:** *zone_congestion_idx / (avg_speed_kmh + 1)*.
- **10-11. dep_hour_sin / cos:** *Cyclical trigonometric time encodings*.

Correlation Matrix with Target (is_delayed)



End-to-End Machine Learning Workflow Pipeline



Algorithms Evaluated & The Baseline Model Trap

The Baseline Model Trap (Class Imbalance)

- **Mandatory Baseline Requirement:** College guidelines require a baseline model to benchmark ML performance.
- **Dummy Classifier (strategy='most_frequent'):** Predicts delayed for 100% of observations.
- **Baseline Validation Results:**
  - **Accuracy: 72.15%**
  - **Precision: 72.15%**
  - **Recall: 100.00%**
  - **ROC-AUC: 0.5000 (Random Guessing!)**
- **Core Takeaway:** A model reporting 72% accuracy on this dataset has achieved NOTHING. Legit ML models MUST beat 72% accuracy and achieve ROC-AUC >> 0.50.

Supervised Algorithm Families

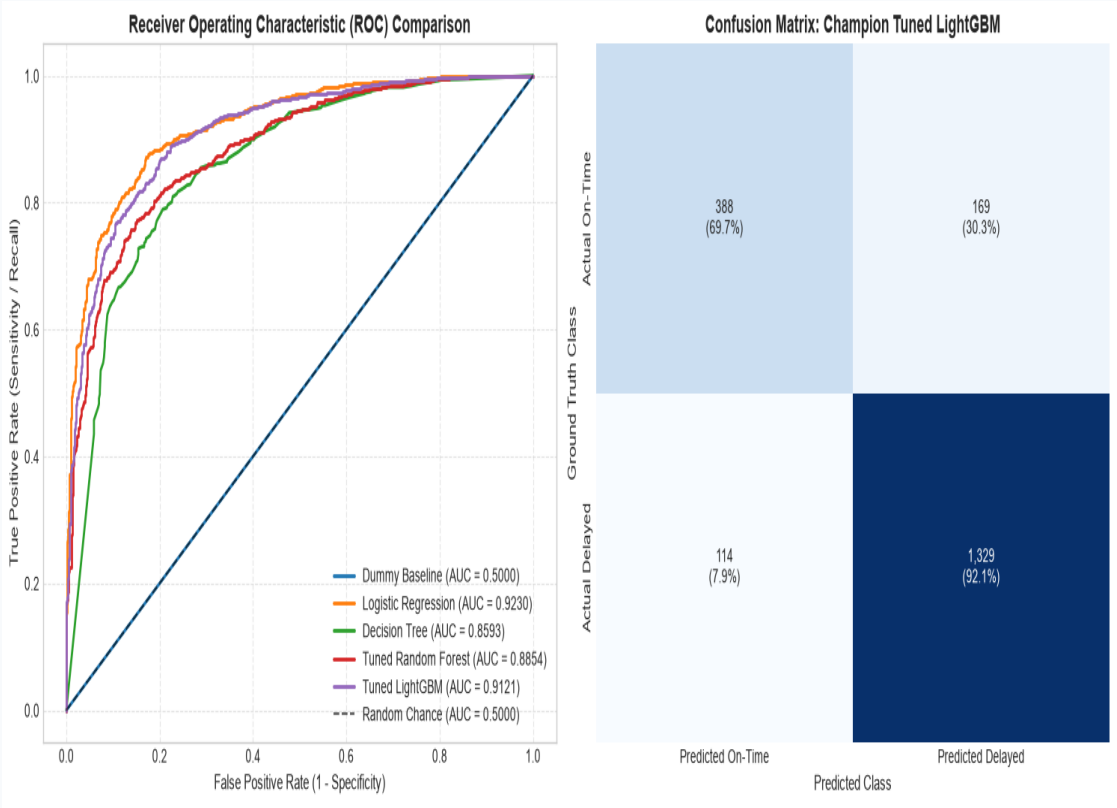
- **1. Logistic Regression:** Linear baseline; fast convergence and strong ranking discrimination.
- **2. Decision Tree:** Non-linear Gini splits; captures interactions but prone to overfitting.
- **3. Random Forest:** Bagged ensemble of de-correlated trees; reduces variance and yields high recall.
- **4. LightGBM (Champion):** Gradient-boosted decision trees using leaf-wise growth; optimal speed & accuracy on tabular data.
- **Hyperparameter Tuning:** 2-fold cross-validation with RandomizedSearchCV optimizing ROC-AUC.

Model Comparison & Champion Model Evaluation

Official Model Benchmark (Validation Set)

Model	Accuracy	Recall	F1	ROC-AUC
Baseline (Dummy)	0.7215	1.0000	0.8382	0.5000
Logistic Regression	0.8570	0.9148	0.9023	0.9229
Decision Tree	0.8130	0.8870	0.8725	0.8593
Random Forest	0.7730	0.9917	0.8631	0.8831
LightGBM	0.8560	0.9148	0.9016	0.9152
Tuned Random Forest	0.7870	0.9854	0.8697	0.8854
Tuned LightGBM ★	**0.8585**	**0.9210**	**0.9038**	**0.9121**

ROC Curves & Confusion Matrix

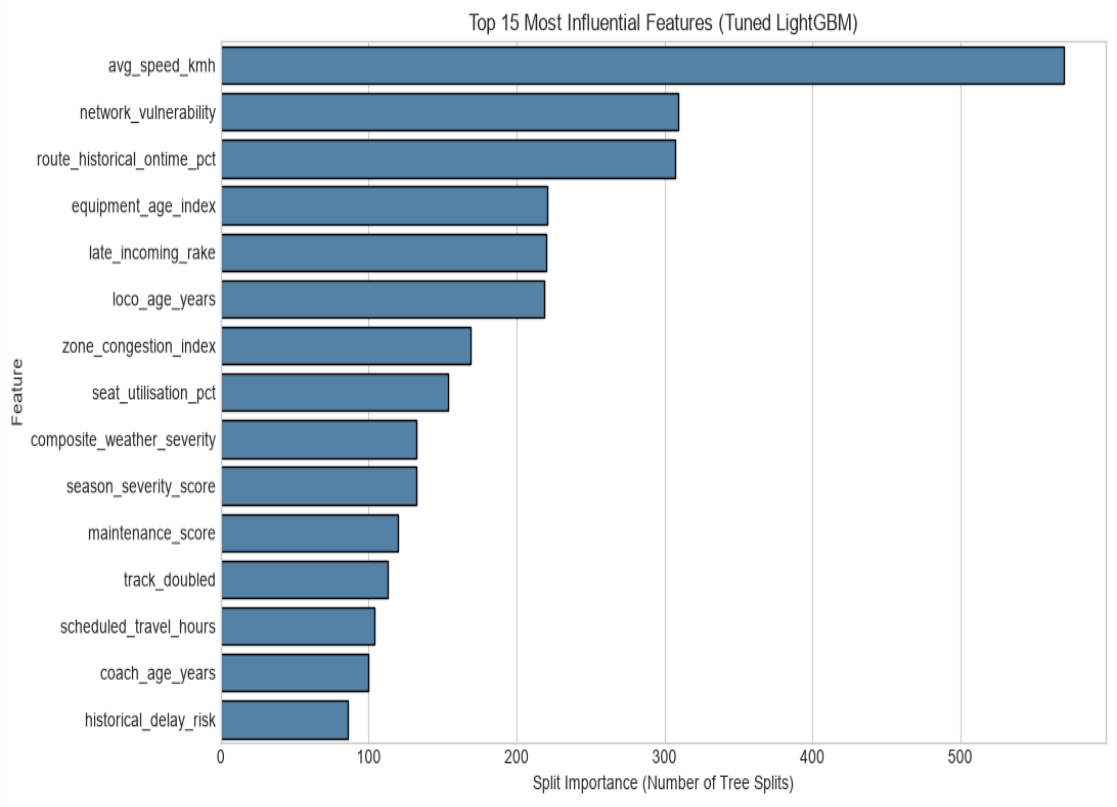


Error Analysis & Model Interpretation

Validation Error Matrix Analysis

- **Confusion Matrix Breakdown (n=2,000):**
  - **True Positives (TP = 1,329):** 92.10% of delayed trains correctly flagged.
  - **True Negatives (TN = 388):** 69.66% of on-time trains correctly cleared.
  - **False Negatives (FN = 114):** Missed Delays (7.90%). Critical dispatch risk.
  - **False Positives (FP = 169):** False Alarms (30.34%). High congestion zones.
- **Feature Importance Highlights:**
  - 5 of the top 15 features are our domain engineered features!
  - **avg_speed_kmh (#1, 570 splits):** Strongest discriminator.
  - **network_vulnerability (#2):** Trunk line corridor exposure.
  - **equipment_age_index (#4):** Rolling stock mechanical wear.

Top 15 Most Influential Features (LightGBM)

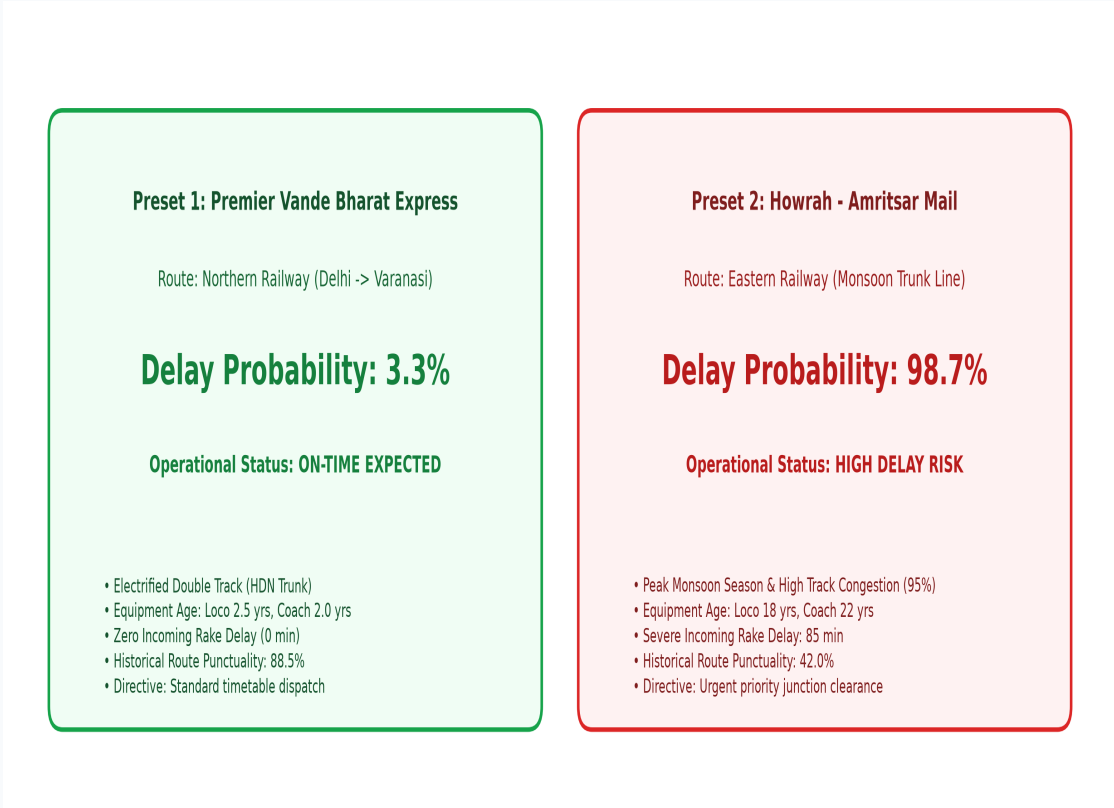


Decision Support Deployment & Live Demonstration

Operational Triage System & Audit

- **Streamlit Web Application (app.py):** Sub-15ms inference using cached LightGBM pipeline.
- **Three Operational Risk Tiers:**
  - **GREEN (<40% Probability):** On-time expected; standard dispatch priority.
  - **AMBER (40%–65% Probability):** Advisory caution; monitor turnaround buffer.
  - **RED (>65% Probability):** Urgent operational intervention & junction clearance.
- **Verified Test Set Inference:** Generated *submission.csv* for all 375,000 test journeys passing all 7 structural checks (0 nulls, 0 duplicate IDs).

Demonstration Scenarios: Vande Bharat vs Mail/Express



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Limitations, Future Enhancements & Conclusion

Limitations & Future Roadmap

- **Hardware-Driven Sample:** Trained on 10,000-record sample to prevent local RAM exhaustion. Roadmap: Distributed PySpark cloud clusters on full 1.5M records.
- **train_number Exclusion:** Excluded to prevent 5,000+ sparse dummy columns. Roadmap: Native CatBoost categorical handling.
- **Static Pre-Departure Scope:** Lacks live GPS block section telemetry. Roadmap: Integrate National Train Enquiry System (NTES) live GPS APIs.
- **Project Conclusion:** Delivered a verified 19-stage ML pipeline. Tuned LightGBM achieved 85.85% accuracy and 92.10% recall, significantly outperforming the 72% dummy baseline.

Academic References & Conclusion

1. Ministry of Railways. (2024). Indian Railways Year Book & Punctuality Stats.
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3. Ke, G., et al. (2017). LightGBM: Efficient Gradient Boosting. NeurIPS 30.
4. Pedregosa, F., et al. (2011). Scikit-learn: ML in Python. JMLR 12.
5. Adani University. (2026). ECSCI24302 Machine Learning Essentials Instructions.

Thank you! We welcome questions from the examination committee.